

```
import cv2
import matplotlib.pyplot as plt
import numpy as np
from sklearn.mixture import GaussianMixture
from skimage.color import rgb2gray
```

```
cap=cv2.VideoCapture('/content/traffic.avi')
```

```
r,f = cap.read()
plt.imshow(f)
```

<matplotlib.image.AxesImage at 0x7da80377b1a0>



```
frames = []
number_of_frames = 0
cap=cv2.VideoCapture('/content/traffic.avi')
while True:
    r,f=cap.read()
    if f is not None:
        frames.append(f)
        number_of_frames+=1
    else:
        break
```

```
print(number_of_frames)
```

```
120
```

```
plt.imshow(frames[119])
```



<matplotlib.image.AxesImage at 0x7da7fb993980>



```
frames=np.array(frames)
cap=cv2.VideoCapture('/content/traffic.avi')
r,f = cap.read()
rows,cols,ch=f.shape
display(rows, cols,ch)
```

```
120
160
3
```

```
gmm=GaussianMixture(n_components=3)
cap=cv2.VideoCapture('/content/traffic.avi')
n=10
background=np.zeros((rows,cols,ch))
for r in range(rows):
    for c in range(cols):
        for k in range(ch):
            temp=frames[0:n,r,c,k]
            temp=temp.reshape(n,1)
            gmm.fit(temp)
            means=gmm.means_
            weights=gmm.weights_
            sequence=np.argmax(weights)
            b=int(means[sequence][0])
            background[r,c,k]=b
```

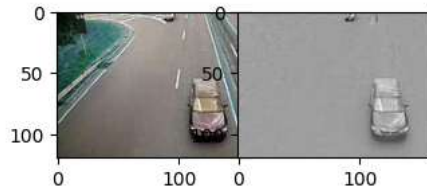
```
background=background/np.max(background)
plt.imshow(background)
```

<matplotlib.image.AxesImage at 0x7da7fb98eea0>



```
plt.subplot(3,1,1)
plt.imshow(frames[50])
plt.subplot(3,2,2)
foreground50=rgb2gray(frames[50])-rgb2gray(background)
plt.imshow(foreground50,cmap='gray')
```

<matplotlib.image.AxesImage at 0x7da7fa9affb0>

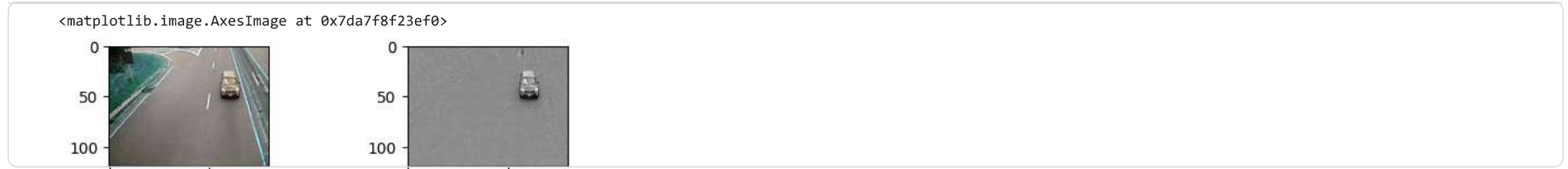


```
plt.subplot(3,2,3)
plt.imshow(frames[80])
plt.subplot(3,2,4)
foreground80=rgb2gray(frames[80])-rgb2gray(background)
plt.imshow(foreground80,cmap='gray')
```

<matplotlib.image.AxesImage at 0x7da7fab423f0>



```
plt.subplot(3,2,5)
plt.imshow(frames[110])
plt.subplot(3,2,6)
foreground110=rgb2gray(frames[110])-rgb2gray(background)
plt.imshow(foreground110,cmap='gray')
```



CONCLUSION

GMM is used to determine the background frame for first 10 frames of the given video .The background subtraction method shows only cars on 50th ,80th and 110th frame .The same can be reperted for all frame to track the movement for car in entire video . All the frame can be converted to video to track the motion of moving cars